
Algorithm 1 Online Adaptation Procedure of AdaPGC+

Require: Incoming minibatch $\mathcal{B}_t = \{(x_i^a, x_i^v)\}_{i=1}^B$; pretrained model f_θ and its EMA copy $f_{\bar{\theta}}$; online Gaussian models $\{\mathcal{G}_a, \mathcal{G}_v, \mathcal{G}_F\}$; cross-view statistics $\{\Sigma^{aF}, \Sigma^{vF}\}$; hyperparameters $T, \beta, \gamma, \lambda_g, \lambda_c$.

Ensure: Predictions $\{\hat{y}_i\}_{i=1}^B$ and updated model/statistics.

- 1: Extract network logits, representations, and availability masks:

$$\{\mathbf{s}_i, \mathbf{z}_i^a, \mathbf{z}_i^v, \mathbf{z}_i^F, m_i\}_{i=1}^B \leftarrow f_\theta(\mathcal{B}_t), m_i \in \{F, a, v\}.$$
- 2: Compute soft assignments $\mathbf{q}_i \leftarrow \text{softmax}(\mathbf{s}_i)$.
- 3: Update the sufficient statistics of each available Gaussian model:

$$\Delta N_c = \sum_i q_{ic}, \quad \Delta \mathbf{S}_c = \sum_i q_{ic} \mathbf{z}_i, \quad \Delta \mathbf{Q}_c = \sum_i q_{ic} \mathbf{z}_i \mathbf{z}_i^\top.$$

- 4: For fully observed samples, update $\Sigma_c^{aF} = \text{Cov}(\mathbf{z}^a, \mathbf{z}^F \mid c)$ and $\Sigma_c^{vF} = \text{Cov}(\mathbf{z}^v, \mathbf{z}^F \mid c)$.
- 5: **for** each sample $i \in \mathcal{B}_t$ **do**
- 6: **if** $m_i = F$ **then**
- 7: Set probabilistic guidance $\mathbf{g}_i \leftarrow \mathcal{G}_F(\mathbf{z}_i^F)$.
- 8: **else**
- 9: Let $m \in \{a, v\}$ be the observed modality.
- 10: **if** paired full-view samples are insufficient **then**
- 11: Use the safe fallback $\mathbf{g}_i \leftarrow \mathcal{G}_m(\mathbf{z}_i^m)$.
- 12: **else**
- 13: Compute observed-view responsibilities:

$$\alpha_{ic}^m \leftarrow \text{softmax}(\mathcal{G}_m(\mathbf{z}_i^m)/T)_c.$$
- 14: Recover a class-conditional fused mean for every class c :

$$\hat{\mathbf{z}}_{ic}^{F|m} \leftarrow \boldsymbol{\mu}_c^F + (\Sigma_c^{mF})^\top (\Sigma_c^m)^{-1} (\mathbf{z}_i^m - \boldsymbol{\mu}_c^m).$$
- 15: Marginalize the recovered fused scores:

$$\mathbf{g}_{i,m \rightarrow F} \leftarrow \sum_{c=1}^C \alpha_{ic}^m \mathcal{G}_F(\hat{\mathbf{z}}_{ic}^{F|m}).$$
- 16: Fuse direct and recovered guidance:

$$\mathbf{g}_i \leftarrow (1 - \beta) \mathbf{g}_{i,m \rightarrow F} + \beta \mathcal{G}_m(\mathbf{z}_i^m).$$
- 17: **end if**
- 18: **end if**
- 19: **end for**
- 20: Compute the probabilistic guidance loss:

$$\mathcal{L}_g \leftarrow -\frac{1}{B} \sum_{i=1}^B \text{softmax}(\mathbf{g}_i)^\top \log \text{softmax}(\mathbf{s}_i).$$

- 21: For each fully observed sample, measure unimodal reliability:

$$D_i^m \leftarrow \text{SKL}(\mathcal{G}_F(\mathbf{z}_i^F), \mathcal{G}_m(\mathbf{z}_i^m)), m \in \{a, v\}.$$
- 22: Select the reliable modality $r_i \leftarrow \arg \min_{m \in \{a, v\}} D_i^m$ and let u_i be the other modality.
- 23: Compute the one-sided alignment loss:

$$\mathcal{L}_c \leftarrow -\frac{1}{|\mathcal{I}_F|} \sum_{i \in \mathcal{I}_F} \log \frac{\exp(\text{sim}(\mathbf{z}_i^{u_i}, \text{sg}(\mathbf{z}_i^{r_i}))/\tau)}{\sum_j \exp(\text{sim}(\mathbf{z}_i^{u_i}, \text{sg}(\mathbf{z}_j^{r_i}))/\tau)}.$$

- 24: Update fusion-attention parameters with $\lambda_g \mathcal{L}_g$; update LayerNorm parameters with $\lambda_c \mathcal{L}_c$; update $\bar{\theta}$ by EMA.
 - 25: Re-evaluate the batch with $f_{\bar{\theta}}$ and predict:

$$\tilde{\mathbf{s}}_i \leftarrow \mathbf{s}_{\bar{\theta}}(x_i) + \gamma \mathbf{g}_i, \hat{y}_i \leftarrow \arg \max_c \tilde{s}_{ic}.$$
 - 26: **return** $\{\hat{y}_i\}_{i=1}^B, \theta, \bar{\theta}$, and the updated online statistics.
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